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Application of the Instrumental Variable Method: Fuzzy Regression Discontinuity

The regression discontinuity method introduced in Chapter 20 and the IV method introduced in Chapters 21–23 are two important examples of *natural experiments*. The study designs are not as ideal as randomized experiments in Part II, but they have features similar to randomized experiments. That's why they are called natural experiments.

Compounding the regression discontinuity method with the IV method yields the *fuzzy regression discontinuity* method, another important natural experiment. I will start with examples and then provide a mathematical formulation.

24.1 Motivating examples

Chapter 20 introduces the regression discontinuity method. The following three examples are slightly different because the treatments received are not deterministic functions of the running variables. Rather, the running variables discontinuously change the probabilities of the treatments received at the cutoff points.

Example 24.1 *In 2000, the Government of India launched the Prime Minister's Village Road Program, and by 2015, this program had funded the construction of all-weather roads to nearly 200,000 villages. Based on village-level data, Asher and Novosad (2020) use the regression discontinuity method to estimate the effect of new feeder roads on various economic variables. The national program guidelines prioritized larger villages according to arbitrary thresholds based on the 2001 Population Census. The treatment variable equals 1 if the village received a new road before the year in which the outcomes were measured. The difference between the population size of a village and the threshold did not determine the treatment variable but affected its probability discontinuously at the cutoff point 0.*

Example 24.2 *Li et al. (2015) used the data on the first-year students enrolled in 2004 to 2006 from two Italian universities to evaluate the causal effect*

of a university grant on the dropout rate. The students were eligible for this grant if their standardized family income was below 15,000 euros. For simplicity, we use the running variable defined as 15,000 minus the standardized family income. To receive this grant, the students must apply first. Therefore, the eligibility and the application status jointly determined the final treatment status. The running variable alone did not determine the treatment status although it changed the treatment probability at the cutoff point 0.

Example 24.3 Amarante et al. (2016) estimated the impact of in-utero exposure to a social assistance program on children's birth outcomes. They used a regression discontinuity induced by the Uruguayan Plan de Atención Nacional a la Emergencia Social. It was a temporary social assistance program targeted to the poorest 10 percent of households, implemented between April 2005 and December 2007. Households with a predicted low-income score below a predetermined threshold were assigned to the program. The predicted income score did not determine whether the mother received at least one program transfer during the pregnancy but it changed the probability of the final treatment received. The birth outcomes included birth weight, weeks of gestation, etc.

The above examples are called fuzzy regression discontinuity in contrast to the (sharp) regression discontinuity in Chapter 20. I will analyze the data in Examples 24.1 and 24.2 in Chapter 24.3 below.

24.2 Mathematical formulation

Let X_i denote the running variable which determines

$$Z_i = I(X_i \geq x_0)$$

with the cutoff point x_0 . The treatment received D_i may not equal Z_i , but $\text{pr}(D_i = 1 \mid X_i = x)$ has a jump at x_0 . Figure 24.1 compares the treatment received probabilities of the sharp regression discontinuity and fuzzy regression discontinuity. It shows a special case of fuzzy regression discontinuity with $\text{pr}(D = 1 \mid X < x_0) = 0$, which is coherent to Example 24.2.

Let Y_i denote the outcome of interest. Viewing Z_i as the treatment assigned, we can define potential outcomes $\{D_i(1), D_i(0), Y_i(1), Y_i(0)\}$. The sharp regression discontinuity of Z allows for the identification of

$$\begin{aligned} \tau_D(x_0) &= E\{D(1) - D(0) \mid X = x_0\} \\ &= \lim_{\varepsilon \rightarrow 0+} E(D \mid Z = 1, X = x_0 + \varepsilon) - \lim_{\varepsilon \rightarrow 0+} E(D \mid Z = 0, X = x_0 - \varepsilon) \end{aligned}$$

and

$$\begin{aligned} \tau_Y(x_0) &= E\{Y(1) - Y(0) \mid X = x_0\} \\ &= \lim_{\varepsilon \rightarrow 0+} E(Y \mid Z = 1, X = x_0 + \varepsilon) - \lim_{\varepsilon \rightarrow 0+} E(Y \mid Z = 0, X = x_0 - \varepsilon) \end{aligned}$$

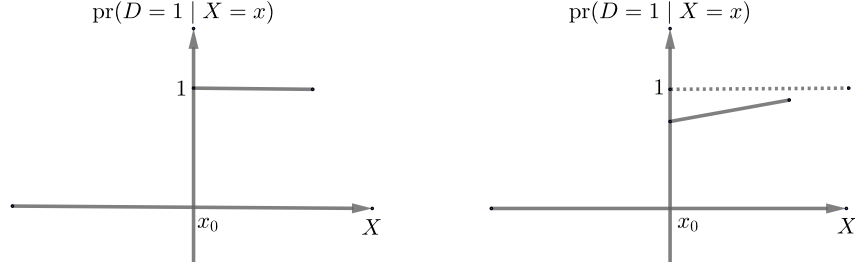


FIGURE 24.1: The treatment assignments of sharp regression discontinuity (left) and fuzzy regression discontinuity (right)

based on Theorem 20.2. Using Z as an IV for D and imposing the IV assumptions at $X = x_0$, we can identify the local complier average causal effect by applying Theorem 21.1.

Theorem 24.1 *Assume that the treatment is determined by $Z = I(X \geq x_0)$ where x_0 is a predetermined threshold. Assume monotonicity*

$$D_i(1) \geq D_i(0)$$

and exclusion restriction

$$D_i(1) = D_i(0) \implies Y_i(1) = Y_i(0)$$

in the infinitesimal neighborhood of x_0 . The local complier average causal effect, defined as

$$\tau_c(x_0) = E\{Y(1) - Y(0) \mid D(1) > D(0), X = x_0\},$$

can be identified by

$$\tau_c(x_0) = \frac{E\{Y(1) - Y(0) \mid X = x_0\}}{E\{D(1) - D(0) \mid X = x_0\}}.$$

Further assume that $E\{D(1) \mid X = x\}$ and $E\{Y(1) \mid X = x\}$ are continuous from the right at $x = x_0$, and $E\{D(0) \mid X = x\}$ and $E\{Y(0) \mid X = x\}$ are continuous from the left at $x = x_0$. The local complier average causal effect can be identified by

$$\tau_c(x_0) = \frac{\lim_{\varepsilon \rightarrow 0+} E(Y \mid Z = 1, X = x_0 + \varepsilon) - \lim_{\varepsilon \rightarrow 0+} E(Y \mid Z = 0, X = x_0 - \varepsilon)}{\lim_{\varepsilon \rightarrow 0+} E(D \mid Z = 1, X = x_0 + \varepsilon) - \lim_{\varepsilon \rightarrow 0+} E(D \mid Z = 0, X = x_0 - \varepsilon)}$$

if $E(D \mid X = x)$ has a non-zero jump at $x = x_0$.

Theorem 24.1 is a superposition of Theorems 20.2 and 21.1. I leave its proof as Problem 24.1.

In both sharp and fuzzy regression discontinuity, the key is to specify the neighborhood around the cutoff point. Practically, a smaller neighborhood leads to a smaller bias but a larger variance, while a larger neighborhood leads to a larger bias but a smaller variance. That is, we face a bias-variance trade-off. Some automatic procedures exist based on some statistical criteria, which rely on some strong conditions. It seems wiser to conduct sensitivity analysis over a range of the choice of the neighborhood.

Assume that we have specified the neighborhood of x_0 determined by a bandwidth h . For data with $X_i \in [x_0 - h, x_0 + h]$, we can estimate $\tau_D(x_0)$ by

$$\hat{\tau}_D(x_0) = \text{the coefficient of } Z_i \text{ in the OLS fit of } D_i \text{ on } \{1, Z_i, R_i, L_i\},$$

and estimate $\tau_Y(x_0)$

$$\hat{\tau}_Y(x_0) = \text{the coefficient of } Z_i \text{ in the OLS fit of } Y_i \text{ on } \{1, Z_i, R_i, L_i\},$$

recalling the definitions $R_i = \max(X_i - x_0, 0)$ and $L_i = \min(X_i - x_0, 0)$. Then we can estimate the local complier average causal effect by

$$\hat{\tau}_c(x_0) = \hat{\tau}_Y(x_0) / \hat{\tau}_D(x_0).$$

This is an indirect least squares estimator. By Theorem 23.1, it is numerically identical to

$$\text{the coefficient of } D_i \text{ in the TSLS fit of } Y_i \text{ on } \{1, D_i, R_i, L_i\}$$

with D_i instrumented by Z_i . In sum, after specifying h , the estimation of $\tau_c(x_0)$ reduces to a TSLS procedure with the local data around the cutoff point.

24.3 Application

24.3.1 Re-analyzing Asher and Novosad (2020)'s data

Revisit Example 24.1. We can compute the point estimates and standard errors for a sequence of h based on the outcome `occupation_index_andrsn`.

```
library("car")
road_dat = read.csv("indianroad.csv")
table(road_dat$t, road_dat$r2012)
road_dat$runv = road_dat$left + road_dat$right
## sensitivity analysis
seq.h = seq(10, 80, 1)
frd_sa = lapply(seq.h, function(h){
  road_sub = subset(road_dat, abs(runv) <= h)
  road_sub$r2012hat = lm(r2012 ~ t + left + right,
```

```

                                data = road_sub)$fitted.values
  tslsreg = lm(occupation_index_andrsn ~ r2012hat + left + right,
              data = road_sub)
  res = with(road_sub,
            {
              occupation_index_andrsn -
              cbind(1, r2012, left, right) %*% coef(tslsreg)
            })
  tslsreg$residuals = as.vector(res)

  c(coef(tslsreg)[2],
    sqrt(hccm(tslsreg, type = "hc2")[2, 2]),
    length(res))
})
frd_sa = do.call(rbind, frd_sa)

```

Figure 24.2 shows the results. The treatment effect is not significant unless h is large.

The package `rdrobust` selects the bandwidth automatically. The results suggest that receiving a new road did not affect the outcome significantly.

```

> library("rdrobust")
> frd_road = with(road_dat,
+               {
+                 rdrobust(y = occupation_index_andrsn,
+                         x = runv,
+                         c = 0,
+                         fuzzy = r2012)
+               })
> res = cbind(frd_road$coef, frd_road$se)
> round(res, 3)

```

	Coeff	Std. Err.
Conventional	-0.253	0.301
Bias-Corrected	-0.283	0.301
Robust	-0.283	0.359

24.3.2 Re-analyzing Li et al. (2015)'s data

Revisit Example 24.2. Recall that the running variable is 15,000 minus the standardized income. In the analysis, I restrict the data to a subset with this running between $[-5,000, 5,000]$, and then divide the running variable by 5,000 so that the running variable is bounded between $[-1, 1]$ at cutoff point zero. We can compute the point estimates and standard errors for a sequence of h .

```

library("car")
italy = read.csv("italy.csv")
italy$left = pmin(italy$rv0, 0)
italy$right = pmax(italy$rv0, 0)

```

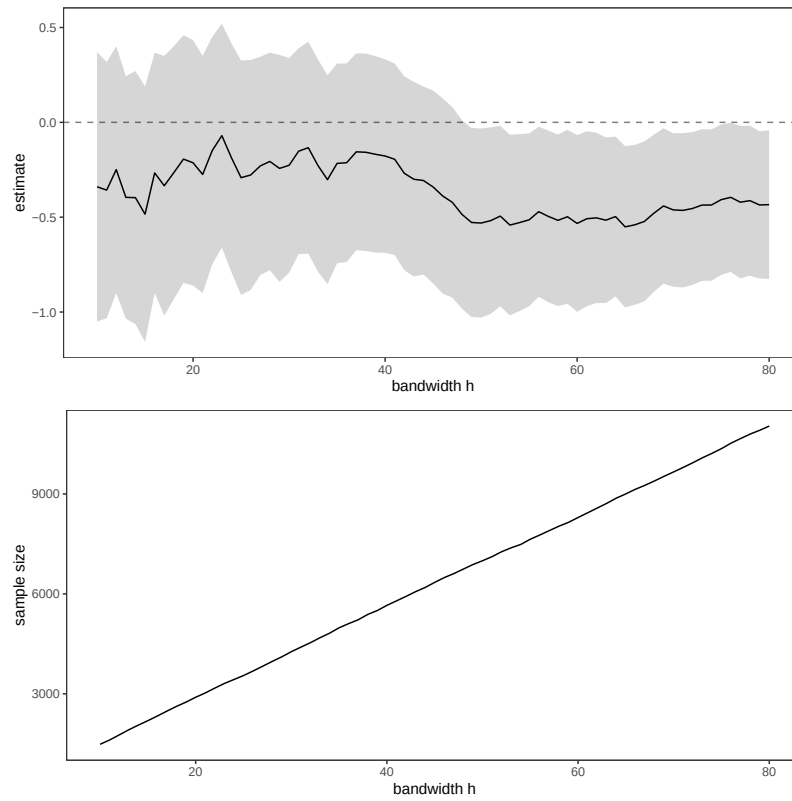


FIGURE 24.2: Re-analyzing Asher and Novosad (2020)'s data, with point estimates and standard errors from TSLS.

```
## sensitivity analysis
seq.h = seq(0.1, 1, 0.01)
frd_sa = lapply(seq.h, function(h){
  italy_sub = subset(italy, abs(rv0)<=h)
  italy_sub$Dhat = lm(D ~ Z + left + right,
                      data = italy_sub)$fitted.values
  tslsreg = lm(outcome ~ Dhat + left + right,
               data = italy_sub)
  res = with(italy_sub,
             {
               outcome -
               cbind(1, D, left, right)%*%coef(tslsreg)
             })
  tslsreg$residuals = as.vector(res)

  c(coef(tslsreg)[2],
    sqrt(hccm(tslsreg, type = "hc2")[2, 2]),
    length(res))
})
frd_sa = do.call(rbind, frd_sa)
```

Figure 24.3 shows the results and suggests that the university grant did not affect the dropout rate significantly.

The results based on the package `rdrobust` reach the same conclusion.

```
> library("rdrobust")
> frd_italy = with(italy,
+                 {
+                   rdrobust(y = outcome,
+                           x = rv0,
+                           c = 0,
+                           fuzzy = D)
+                 })
> res = cbind(frd_italy$coef, frd_italy$sse)
> round(res, 3)
```

	Coeff	Std. Err.
Conventional	-0.149	0.101
Bias-Corrected	-0.155	0.101
Robust	-0.155	0.121

24.4 Discussion

Both Chapter 20 and this chapter formulate regression discontinuity based on the continuity of the conditional expectations of the potential outcomes given the running variables. This perspective is mathematically simpler but it only

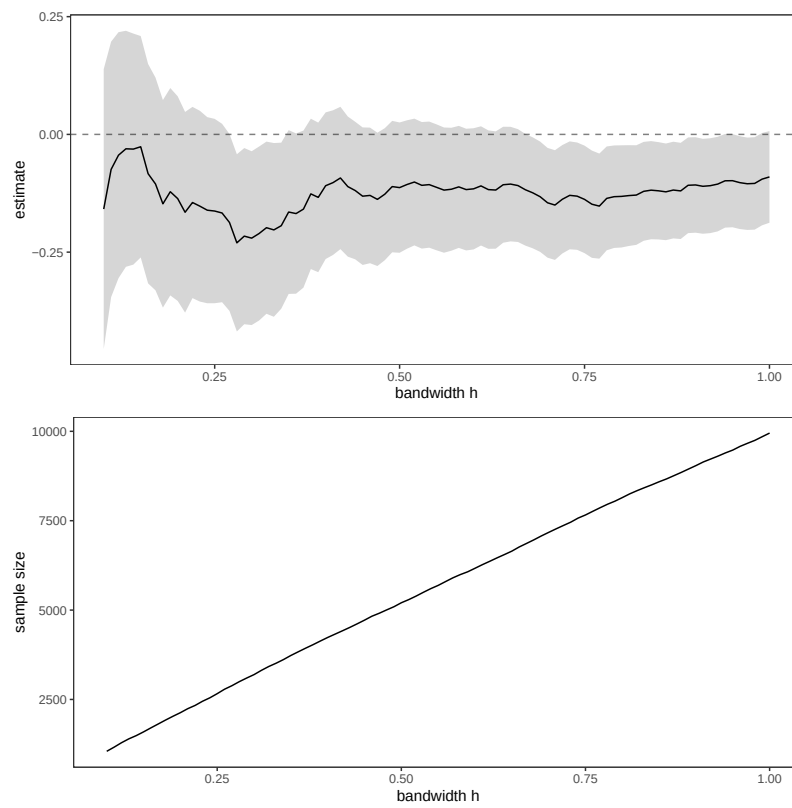


FIGURE 24.3: Re-analyzing Li et al. (2015)'s data, with point estimates and standard errors from TSLS.

identifies the local effects precisely at the cutoff point of the running variable. Hahn et al. (2001) started this line of literature.

An alternative, not so dominant perspective is based on *local randomization* (Cattaneo et al., 2015; Li et al., 2015). If we view the running variable as a noisy measure of some underlying truth and the cutoff point is somewhat arbitrarily chosen, the units near the cutoff point do not differ systematically. This suggests that in a small neighborhood of the cutoff point, the units receive the treatment and the control in a random fashion just as in a randomized experiment. Similar to the issue of choosing h in the first perspective, it is crucial to decide how local should the randomized experiment be under the regression discontinuity. It is not easy to quantify the intuition mathematically, and again conducting sensitivity analysis with a range of h seems a reasonable approach in the second perspective as well.

See Sekhon and Titiunik (2017) for a more conceptual discussion of regression discontinuity.

24.5 Homework Problems

24.1 Proof of Theorem 24.1

Prove Theorem 24.1.

24.2 Data analysis

Section 24.3.1 reports the estimate of the effect on `occupation_index_andrsn`. Four other outcome variables are `transport_index_andrsn`, `firms_index_andrsn`, `consumption_index_andrsn`, and `agriculture_index_andrsn`, with meanings defined in the original paper. Estimate the effects on these outcomes.

24.3 Reflection on the analysis of Li et al. (2015)'s data

In Li et al. (2015), a key variable determining the treatment status is the binary application status A , which has potential outcomes $A(1)$ and $A(0)$ corresponding to the treatment $Z = 1$ and control $Z = 0$. By definition,

$$D(1) = A(1), \quad D(0) = 0,$$

so the compliers $\{D(1), D(0)\} = (1, 0)$ is equivalent to $A(1) = 1$. So

$$\tau_c(x_0) = E\{Y(1) - Y(0) \mid A(1) = 1, X = x_0\}.$$

Section 24.3.2 used the whole data set to estimate $\tau_c(x_0)$.

An alternative analysis is to use the units with $A = 1$ only. Then the treatment status is determined by X . However, this analysis can be problematic

because

$$\begin{aligned} & \lim_{\varepsilon \rightarrow 0+} E\{Y \mid A = 1, X = x_0 + \varepsilon\} - \lim_{\varepsilon \rightarrow 0+} E\{Y \mid A = 1, X = x_0 - \varepsilon\} \\ = & E\{Y(1) \mid A(1) = 1, X = x_0\} - E\{Y(0) \mid A(0) = 1, X = x_0\}. \end{aligned} \quad (24.1)$$

Prove (24.1) and explain why this analysis can be problematic.

Remark: The left-hand side of (24.1) is the identification formula of the local average treatment effect at $X = x_0$, conditional on $A = 1$. The right-hand side of (24.1) is the difference in means of the potential outcomes for the subgroups of units with $(A(1) = 1, X = x_0)$ and $(A(0) = 1, X = x_0)$, respectively. See Chapter 26 later for related discussions.

24.4 *Recommended reading*

Imbens and Lemieux (2008) gave practical guidance to regression discontinuity based on the potential outcomes framework. Lee and Lemieux (2010) reviewed regression discontinuity and its applications in economics.

Application of the Instrumental Variable Method: Mendelian Randomization

Katan (1986) was concerned with the observational studies suggesting that low serum cholesterol levels were associated with the risk of cancer. As we have discussed, however, observational studies suffer from unmeasured confounding. Consequently, it is difficult to interpret the observed association as causality. In the particular problem studied by Katan (1986), it is even possible that early stages of cancer reversely cause low serum cholesterol levels. Disentangling the causal effect of the serum cholesterol level on cancer seems a hard problem using standard epidemiologic studies. Katan (1986) argued that Apolipoprotein E genes are associated with serum cholesterol levels but do not directly affect the cancer status. So if low serum cholesterol levels cause cancer, we should observe differences in cancer risks among people with and without the genotype that leads to different serum cholesterol levels. Using our language for causal inference, Katan (1986) proposed to use Apolipoprotein E genes as IVs.

Katan (1986) did not conduct any data analysis but just proposed a conceptual design that could address not only *unmeasured confounding* but also *reverse causality*. Since then, more complicated and sophisticated studies have been conducted thanks to the modern genome-wide association studies. These studies used genetic information as IVs for exposures in epidemiologic studies to estimate the causal effects of exposures on outcomes. They were all motivated by *Mendel's second law*, the law of *random assortment*, which suggests that the inheritance of one trait is independent of the inheritance of other traits. Therefore, the method of using genetic information as IV is called *Mendelian Randomization* (MR).

25.1 Background and motivation

Graphically, Figure 25.1 shows the causal diagram on the treatment D , outcome Y , unmeasured confounder U , as well as the genetic IVs $G_1, \dots, G_p$. In many MR studies, the genetic IVs are single nucleotide polymorphisms

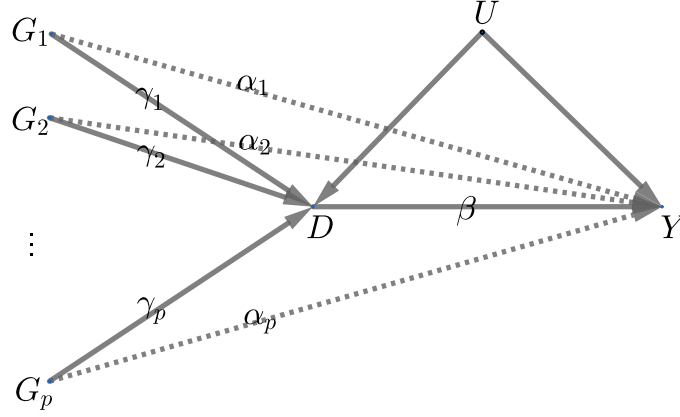


FIGURE 25.1: Causal graph for MR

(SNPs). Because of pleiotropy,¹ it is possible that the genetic IVs have a direct effect on the outcome of interest, so Figure 25.1 also allows for the violation of the exclusion restriction assumption.

The standard linear IV model assumes away the direct effect of the IVs on the outcome. Definition 25.1 below gives both the structural and reduced forms.

Definition 25.1 (linear IV model) *The standard linear IV model*

$$Y = \beta_0 + \beta D + \beta_u U + \varepsilon_Y, \quad (25.1)$$

$$D = \gamma_0 + \gamma_1 G_1 + \cdots + \gamma_p G_p + \gamma_u U + \varepsilon_D, \quad (25.2)$$

has reduced form

$$Y = \beta_0 + \beta\gamma_0 + \beta\gamma_1 G_1 + \cdots + \beta\gamma_p G_p + (\beta_u + \beta_0\gamma_u)U + \varepsilon_Y, \quad (25.3)$$

$$D = \gamma_0 + \gamma_1 G_1 + \cdots + \gamma_p G_p + \gamma_u U + \varepsilon_D, \quad (25.4)$$

Definition 25.2 below allows for the violation of exclusion restriction. Then, $G_1, \dots, G_p$ are not valid IVs.

Definition 25.2 (linear model with possibly invalid IVs) *The linear model*

$$Y = \beta_0 + \beta D + \alpha_1 G_1 + \cdots + \alpha_p G_p + \beta_u U + \varepsilon_Y, \quad (25.5)$$

$$D = \gamma_0 + \gamma_1 G_1 + \cdots + \gamma_p G_p + \gamma_u U + \varepsilon_D, \quad (25.6)$$

¹Pleiotropy occurs when one gene influences two or more seemingly unrelated phenotypic traits.

has reduced form

$$Y = (\beta_0 + \beta\gamma_0) + (\alpha_1 + \beta\gamma_1)G_1 + \cdots + (\alpha_p + \beta\gamma_p)G_p + (\beta_u + \beta\gamma_u)U + \varepsilon_Y, \quad (25.7)$$

$$D = \gamma_0 + \gamma_1 G_1 + \cdots + \gamma_p G_p + \gamma_u U + \varepsilon_D. \quad (25.8)$$

Definitions 25.1 and 25.2 are slightly different from the linear IV model in Chapter 23. They include the confounder U explicitly. But this slight difference does not change the discussion fundamentally.

In Definition 25.1 with exclusion restriction, we have

$$\Gamma_j = \beta\gamma_j, \quad (j = 1, \dots, p).$$

In Definition 25.2 without exclusion restriction, we have

$$\Gamma_j = \alpha_j + \beta\gamma_j, \quad (j = 1, \dots, p).$$

If we have individual data, we can apply the classic TSLS estimator to estimate β under the linear IV model in Definition 25.1. However, most MR studies do not have individual data but rather summary statistics from multiple genome-wide association studies. A canonical MR study with summary statistics has the following data structure.

Assumption 25.1 (MR study with summary statistics) (a) We have the regression coefficients of the treatment on the genetic IVs $\hat{\gamma}_1, \dots, \hat{\gamma}_p$ as well as the standard errors $\text{se}_{D1}, \dots, \text{se}_{Dp}$. Assume

$$\hat{\gamma}_1 \rightarrow \gamma_1, \dots, \hat{\gamma}_p \rightarrow \gamma_p \quad (25.9)$$

in probability, and ignore the uncertainty in the standard errors.

(b) We have the regression coefficients of the outcome on the genetic IVs $\hat{\Gamma}_1, \dots, \hat{\Gamma}_p$ as well as with standard errors $\text{se}_{Y1}, \dots, \text{se}_{Yp}$. Assume

$$\hat{\Gamma}_1 \rightarrow \Gamma_1, \dots, \hat{\Gamma}_p \rightarrow \Gamma_p \quad (25.10)$$

in probability, and ignore the uncertainty in the standard errors.

(c) Assume $\hat{\gamma}_1, \dots, \hat{\gamma}_p, \hat{\Gamma}_1, \dots, \hat{\Gamma}_p$ are jointly Normal and independent.

The asymptotic Normality of the regression coefficients can be justified by the CLT. The standard errors are accurate estimates of the true standard errors in large samples. Therefore, the only subtle assumption is the joint independence of the regression coefficients. The independence of the $\hat{\gamma}_j$'s and the $\hat{\Gamma}_j$'s are reasonable because they are often calculated based on different samples.

The independence among the $\hat{\gamma}_j$'s can be reasonable if the G_j 's are independent and the true linear model for D holds with homoskedastic error terms. See Chapter B.4. However, this assumption can be tricky if the error

term of the linear model is heteroskedastic. Without the independence of the G_j 's, it is hard to justify the independence. If the regression coefficients are from nonlinear models, then it is even harder to justify the independence. The independence among the $\hat{\Gamma}_j$'s follows from a similar argument.

This chapter focuses on the statistical inference of β based on the above summary statistics in Assumption 25.1.

25.2 MR based on summary statistics

25.2.1 Fixed-effect estimator

Under Definition 25.1, $\alpha_j = 0$ which implies that $\beta = \Gamma_j/\gamma_j$ for all j . A simple approach is based on the so-called *meta-analysis* (Bowden et al., 2018), that is, combining multiple estimates $\hat{\beta}_j = \hat{\Gamma}_j/\hat{\gamma}_j$ for the common parameter β , also called the fixed effect. Using delta method (see Example A.3), the ratio estimator $\hat{\beta}_j$ has approximate squared standard error

$$\text{se}_j^2 = (\text{se}_{Yj}^2 + \hat{\beta}_j^2 \text{se}_{Dj}^2)/\hat{\gamma}_j^2.$$

Therefore, the best linear combination to estimate β is the Fisher weighting based on the inverses of the variances (see Problem A.6):

$$\hat{\beta}_{\text{fisher0}} = \frac{\sum_{j=1}^p \hat{\beta}_j / \text{se}_j^2}{\sum_{j=1}^p 1 / \text{se}_j^2}$$

which has variance $(\sum_{j=1}^p 1/\text{se}_j^2)^{-1}$. Ignoring the uncertainty due to $\hat{\gamma}_j$ quantified by se_{Dj} , the estimator reduces to

$$\hat{\beta}_{\text{fisher1}} = \frac{\sum_{j=1}^p \hat{\beta}_j \hat{\gamma}_j^2 / \text{se}_{Yj}^2}{\sum_{j=1}^p \hat{\gamma}_j^2 / \text{se}_{Yj}^2} = \frac{\sum_{j=1}^p \hat{\Gamma}_j \hat{\gamma}_j / \text{se}_{Yj}^2}{\sum_{j=1}^p \hat{\gamma}_j^2 / \text{se}_{Yj}^2},$$

which has variance $(\sum_{j=1}^p \hat{\gamma}_j^2 / \text{se}_{Yj}^2)^{-1}$. Inference based on $\hat{\beta}_{\text{fisher1}}$ is suboptimal although it is more widely used in practice (Bowden et al., 2018). Both $\hat{\beta}_{\text{fisher0}}$ and $\hat{\beta}_{\text{fisher1}}$ are called the *fixed-effect estimators*.

Focus on the suboptimal yet simpler estimator $\hat{\beta}_{\text{fisher1}}$. Under Definition 25.2, we can show that

$$\hat{\beta}_{\text{fisher1}} \rightarrow \frac{\sum_{j=1}^p \Gamma_j \gamma_j / \text{se}_{Yj}^2}{\sum_{j=1}^p \gamma_j^2 / \text{se}_{Yj}^2} = \beta + \frac{\sum_{j=1}^p \alpha_j \gamma_j / \text{se}_{Yj}^2}{\sum_{j=1}^p \gamma_j^2 / \text{se}_{Yj}^2}$$

in probability. If $\alpha_j = 0$ for all j , $\hat{\beta}_{\text{fisher1}}$ is consistent. Even if this does not hold, it is still possible that $\hat{\beta}_{\text{fisher1}}$ is consistent as long as the inner product between α_j and γ_j weighted by $1/\text{se}_{Yj}^2$ is zero. This holds if we have many genetic instruments and violation of the exclusion restriction, captured by α_j , is an independent random draw from a distribution with mean zero.

25.2.2 Egger regression

Start with Definition 25.1. With the true parameters, we have

$$\Gamma_j = \beta\gamma_j \quad (j = 1, \dots, p);$$

with the estimates, the above identity holds only approximately

$$\hat{\Gamma}_j \approx \beta\hat{\gamma}_j \quad (j = 1, \dots, p).$$

This seems a classic least-squares problem of $\{\hat{\Gamma}_j\}_{j=1}^p$ on $\{\hat{\gamma}_j\}_{j=1}^p$. We can fit a WLS of $\hat{\Gamma}_j$ on $\hat{\gamma}_j$, with or without an intercept, possibly weighted by w_j , to estimate β . The following results hold thanks to the algebraic properties of the WLS reviewed in Chapter B.5.

Without an intercept, the coefficient of $\hat{\gamma}_j$ is

$$\hat{\beta}_{\text{egger1}} = \frac{\sum_{j=1}^p \hat{\gamma}_j \hat{\Gamma}_j w_j}{\sum_{j=1}^p \hat{\gamma}_j^2 w_j},$$

which reduces to $\hat{\beta}_{\text{fisher1}}$ if $w_j = 1/\text{se}_{Yj}^2$. The WLS is called the *Egger regression*. It is more general than the fixed-effect estimators in Section 25.2.1. With an intercept, the coefficient of $\hat{\gamma}_j$ is

$$\hat{\beta}_{\text{egger0}} = \frac{\sum_{j=1}^p (\hat{\gamma}_j - \hat{\gamma}_w)(\hat{\Gamma}_j - \hat{\Gamma}_w)w_j}{\sum_{j=1}^p (\hat{\gamma}_j - \hat{\gamma}_w)^2 w_j}$$

where $\hat{\gamma}_w = \sum_{j=1}^p \hat{\gamma}_j w_j / \sum_{j=1}^p w_j$ and $\hat{\Gamma}_w = \sum_{j=1}^p \hat{\Gamma}_j w_j / \sum_{j=1}^p w_j$ are the weighted averages of the $\hat{\gamma}_j$'s and $\hat{\Gamma}_j$'s, respectively.

Even without assuming that all γ_j 's are zero under Definition 25.2, we have

$$\hat{\beta}_{\text{egger0}} \rightarrow \frac{\sum_{j=1}^p (\gamma_j - \gamma_w)(\Gamma_j - \Gamma_w)w_j}{\sum_{j=1}^p (\gamma_j - \gamma_w)^2 w_j} = \beta + \frac{\sum_{j=1}^p (\gamma_j - \gamma_w)(\alpha_j - \alpha_w)w_j}{\sum_{j=1}^p (\gamma_j - \gamma_w)^2 w_j}$$

in probability, where γ_w, Γ_w and α_w are the corresponding weighted averages of the true parameters. So $\hat{\beta}_{\text{egger0}}$ is consistent for β as long as the WLS coefficient of α_j on γ_j is zero. This is weaker than $\alpha_j = 0$ for all j . This weaker assumption holds if γ_j and α_j are realizations of independent random variables, which is called the Instrument Strength Independent of Direct Effect (InSIDE) assumption (Bowden et al., 2015). More interestingly, the intercept from the Egger regression is

$$\hat{\alpha}_{\text{egger0}} = \hat{\Gamma}_w - \hat{\beta}_{\text{egger0}}\hat{\gamma}_w,$$

which, under the InSIDE assumption, converges to

$$\Gamma_w - \beta\gamma_w = \alpha_w$$

in probability. So the intercept estimates the weighted average of the direct effects.

25.3 An example

First, the following function can implement Fisher weighting.

```
fisher.weight = function(est, se)
{
  n = sum(est/se^2)
  d = sum(1/se^2)
  res = c(n/d, sqrt(1/d))
  names(res) = c("est", "se")
  res
}
```

I use the `bmi.sbp` data in the `mr.raps` package (Zhao et al., 2020) to illustrate the Fisher weighting based on different variance estimates. The results based on $\hat{\beta}_{\text{fisher0}}$ and $\hat{\beta}_{\text{fisher1}}$ are similar.

```
> bmisbp = read.csv("mr_bmisbp.csv")
> bmisbp$iv      = with(bmisbp, beta.outcome/beta.exposure)
> bmisbp$se.iv   = with(bmisbp, se.outcome/beta.exposure)
> bmisbp$se.iv1  = with(bmisbp,
+                        sqrt(se.outcome^2 + iv^2*se.exposure^2)/beta.exposure)
> fisher.weight(bmisbp$iv, bmisbp$se.iv)
      est      se
0.31727680 0.05388827
> fisher.weight(bmisbp$iv, bmisbp$se.iv1)
      est      se
0.31576007 0.05893783
```

The Egger regressions with or without the intercept also give very similar results. But the standard errors are quite different from those based on Fisher weighting.

```
> mr.egger = lm(beta.outcome ~ 0 + beta.exposure,
+               data = bmisbp,
+               weights = 1/se.outcome^2)
> summary(mr.egger)$coef
              Estimate Std. Error  t value    Pr(>|t|)
beta.exposure 0.3172768   0.1105994  2.868704 0.004681659
>
> mr.egger.w = lm(beta.outcome ~ beta.exposure,
+                 data = bmisbp,
+                 weights = 1/se.outcome^2)
> summary(mr.egger.w)$coef
              Estimate Std. Error  t value    Pr(>|t|)
(Intercept)  0.0001133328 0.002079418  0.05450217 0.956603931
beta.exposure 0.3172989306 0.110948506  2.85987564 0.004811017
```

Figure 25.2 shows the raw data as well as the fitted Egger regression line with the intercept.

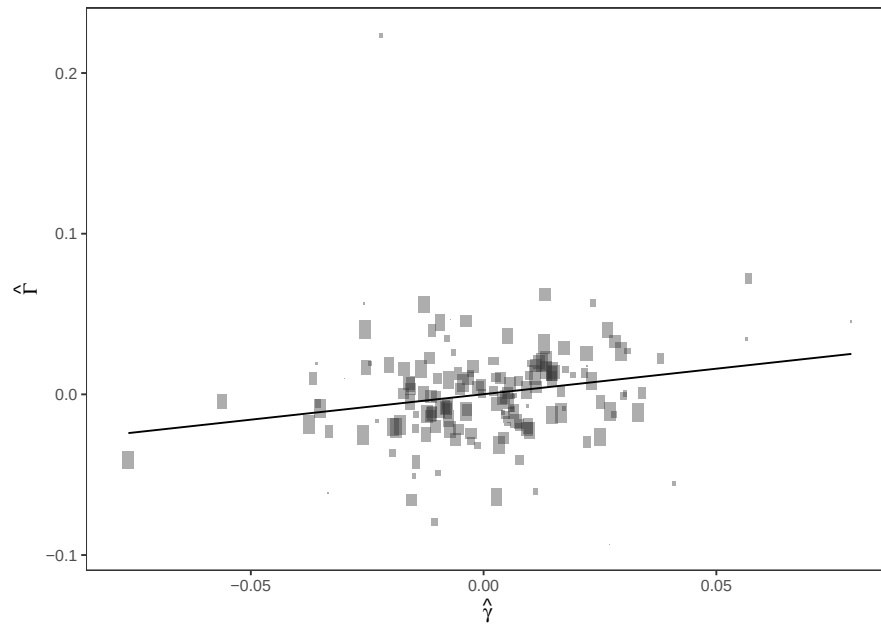


FIGURE 25.2: Scatter plot proportional to the inverse of the variance, with the Egger regression line

25.4 Critiques of the analysis based on MR

MR is an application of the idea of IV. It relies on strong assumptions. I provide three sets of critiques from the conceptual, biological, and technical perspectives.

Conceptually, the treatments in most studies based on MR are not well defined from the potential outcomes perspective. For instance, the treatments are often defined as the cholesterol level or the BMI. They are composite variables and can correspond to complex, non-unique definitions of the hypothetical experiments. The SUTVA often does not hold for these treatments. Recall the discussion in Chapter 2.

Biologically, the fundamental assumptions for the IV analysis may not hold. Mendel's second law ensures that the inheritances of different traits are independent. However, it does not ensure that the candidate IVs are independent of the hidden confounders between the treatment and the outcome. It is possible that these IVs have direct effects on the confounders. It is also possible that some unmeasured genes affect both the IVs and the confounders. Mendel's second law does not ensure the exclusion restriction assumption either. It is possible that the IVs have other causal pathways to the outcome, beyond the pathway through the treatment of interest.

Technically, the statistical assumptions for MR are quite strong. Clearly, the linear IV model is a strong modeling assumption. The independence of the $\hat{\gamma}_j$'s and the $\hat{\Gamma}_j$'s is also quite strong. Other issues in the data-collecting process can further complicate the interpretation of the IV assumptions. For instance, the treatments and outcomes are often measured with errors, and the genome-wide associate studies are often based on the case-control design with the samples conditional on the outcomes (see Chapter B.6.3).

VanderWeele et al. (2014) is an excellent review paper that discusses the methodological challenges in MR.

25.5 Homework Problems

25.1 Data analysis

Analyze the `bmi.bmi` data in the R package `mr.raps`. See the package and Zhao et al. (2020, Section 7.2) for more details.

25.2 Recommended reading

Davey Smith and Ebrahim (2003) reviewed the potentials and limitations of MR.